

Bird Survey

Swadini, Blyde River Canyon

Mpumalanga · South Africa

Seasonal observations 2019 – 2025

95

SPECIES

806

RECORDS

7

SEASONS

42

FAMILIES

Observer

Andries de Vries · Citizen Science · Observation.org

Mixed savannah and riverine forest habitat · ~1.1 km² study area

Western bank, Blyde River below Blyderivierspoort Dam

1. Introduction & Study Context

This report summarises seven consecutive seasons of bird (Aves) observations made by citizen scientist Andries de Vries at Swadini, a Forever Resort situated within the Blyderivierspoort Nature Reserve, Mpumalanga, South Africa. The study area covers the western bank of the Blyde River below the Blyderivierspoort Dam, within the Blyde River Canyon, and encompasses approximately 1.1 km² of mixed habitat.

The habitat combines open savannah and bushveld thickets on the canyon slopes with a ribbon of dense indigenous riverine forest along the Blyde River itself. This juxtaposition supports a characteristically diverse bird community: savannah generalists, forest specialists, waterbirds, and raptors all occur within the same compact study area. The canyon walls attract cliff-nesting eagles while the river corridor provides year-round habitat for kingfishers, the African Finfoot, and Pel's Fishing Owl.

Bird observations were recorded as a component of broader biodiversity surveys focused primarily on Odonata (dragonflies and damselflies). From 2019 to 2024, birds were logged incidentally by survey team members; from 2025, the primary observer adopted systematic weekly recording using the Observation.org mobile app, enabling photo-free submissions and dramatically increasing both effort and species detection. All records from 2019–2024 carry Accepted (with evidence) validation status; 2025 records were uploaded directly to Observation.org.

Table 1 — Season Summary

Year	Records	Species	Survey window	Notes
2019	65	47	May–Nov	Two visits: May–Aug & Nov. Incidental recording.
2020	27	23	Aug–Oct	COVID-19 lockdown; season started in August only.
2021	43	35	Apr–Aug	Incidental; other team members focused on birds.
2022	35	25	Apr–Aug	Incidental recording alongside Odonata surveys.
2023	52	38	Apr–Aug	Incidental; subjective increase in bird activity noted.
2024	76	46	Apr–Jul	Increased awareness; drought conditions observed.
2025	508	67	May–Aug	Systematic weekly recording; mobile app; 12-day gap wk 28 (family emergency).

Species counts are normalised to species level — subspecies records rolled up to parent species. The 2025 count of 67 species is the most complete single-season estimate.

2. Species Richness & Taxonomic Diversity

A total of **95 unique species** across **42 families** were recorded across all seven seasons (806 records). Annual richness ranged from 23 species (2020, lockdown-shortened season) to 67 species (2025, systematic effort). The cumulative species total of 95 substantially exceeds any single-season count, reflecting the value of multi-year presence at a consistent site.

The dominant families by species count are Muscicapidae (Chats & Flycatchers, 7 species in 2025), Estrildidae (Waxbills & Firefinches, 6 species), Nectariniidae (Sunbirds, 5 species), and Accipitridae (Eagles & Hawks, 4 species). The presence of Accipitridae is notable: Verreaux's Eagle, African Goshawk, African Fish Eagle,

African Harrier-Hawk, and Crowned Eagle have all been recorded, reflecting the canyon's exceptional raptor habitat.

Only **4 species were recorded in all 7 seasons**: Trumpeter Hornbill, Red-headed Weaver, Retz's Helmetshrike, and Cape White-eye. This low core count reflects the incidental and low-effort nature of pre-2025 recording rather than genuine scarcity. A further 6 species were recorded in 6 of 7 seasons (African Goshawk, African Pied Wagtail, Ashy Flycatcher, Black-backed Puffback, African Finfoot, and Motacilla aguimp), forming a wider group of near-resident species.

Table 2 — Top 10 Families by Cumulative Species Count

Family	Common name	Species (total)	2025
Muscicapidae	Chats & Flycatchers	8	7
Estrildidae	Waxbills & Firefinches	8	6
Malaconotidae	Bushshrikes & Tchagras	7	3
Nectariniidae	Sunbirds	7	5
Accipitridae	Eagles, Hawks & Kites	6	4
Alcedinidae	Kingfishers	5	3
Cisticolidae	Cisticolas & Allies	4	2
Pycnonotidae	Bulbuls & Greenbuls	4	3
Ploceidae	Weavers	4	3
Lybiidae	African Barbets	4	2

3. Core Resident & Notable Species

The four species recorded in every season regardless of survey effort or duration represent the most reliably detectable components of the Swadini bird community. Trumpeter Hornbill and Retz's Helmetshrike are characteristically noisy, conspicuous species of riverine forest whose calls carry widely. Red-headed Weaver and Cape White-eye are ubiquitous in the savannah-forest edge habitat.

Species	Common name	Family	Seasons
<i>Bycanistes bucinator</i>	Trumpeter Hornbill	Bucerotidae (Hornbills)	7/7
<i>Anaplectes rubriceps</i>	Red-headed Weaver	Ploceidae (Weavers)	7/7
<i>Prionops retzii</i>	Retz's Helmetshrike	Vangidae (Helmetshrikes)	7/7
<i>Zosterops virens</i>	Cape White-eye	Zosteropidae (White-eyes)	7/7
<i>Podica senegalensis</i>	African Finfoot	Heliornithidae (Finfoots)	6/7
<i>Fraseria caerulescens</i>	Ashy Flycatcher	Muscicapidae (Chats & Flycatchers)	6/7
<i>Motacilla aguimp</i>	African Pied Wagtail	Motacillidae (Wagtails)	6/7
<i>Dryoscopus cubla</i>	Black-backed Puffback	Malaconotidae (Bushshrikes)	6/7
<i>Aerospiza tachiro</i>	African Goshawk	Accipitridae (Eagles & Hawks)	6/7

Notable Records

Pel's Fishing Owl (*Scotopelia peli*) — Recorded week 32, 2025. A large, localised owl of intact riverine forest; its presence confirms the quality of the Blyde River corridor within the study area. One of the most sought-after birds in southern Africa.

Crowned Eagle (*Stephanoaetus coronatus*) — Recorded in 2021, 2022 and 2023. Africa's most powerful eagle, requiring large territories and intact forest. Its multi-season occurrence confirms that the canyon holds sufficient habitat and prey for this apex predator.

Narina Trogon (*Apaloderma narina*) — Recorded in 5 seasons. A regularly occurring forest species, confirming the canopy quality of the riverine forest along the study area.

African Finfoot (*Podica senegalensis*) — Recorded in 6 of 7 seasons. A secretive, under-recorded waterbird that is a reliable indicator of undisturbed riverine habitat. Its consistent presence is ecologically significant.

4. Seasonal Phenology

The survey window spans approximately late April to late August (IOC weeks 17–35), covering the latter part of autumn, full winter, and the start of the early-spring warm-up in the Southern Hemisphere. This window does not capture the austral summer (November–March), which is likely the period of maximum bird activity and breeding for many resident species.

Within the observed window, species richness peaks in May and July–August in most years. May captures late-summer resident species still present from the breeding season; July and August reflect the resident winter community at its most stable. The 2025 data, with consistent weekly effort, shows richness maintained throughout May–August (44–54 species per month), suggesting the site supports a robust year-round bird community.

The comparable weeks analysis (IOC weeks surveyed in ≥5 of 7 seasons) confirms the 2025 spike is effort-driven: weeks 19–22 averaged 1–4 species in 2019–2024 but 15–21 species in 2025 under systematic effort. Week 22 in 2019 (21 species) is a notable exception — an intensive single-day session that produced unusually high numbers for that period.

Table 3 — Species Richness by Month and Year

Year	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
2019	–	23	5	22	5	–	–	4
2020	–	–	–	–	11	7	8	–
2021	1	2	12	15	10	–	–	–
2022	1	3	8	15	5	–	–	–
2023	2	8	6	15	15	–	–	–
2024	4	22	17	24	–	–	–	–
2025	–	44	46	36	54	–	–	–

– = month not surveyed that year. 2025 figures reflect systematic weekly effort; earlier years reflect incidental recording.

5. Methodology, Effort & Outlook

Survey method evolution. The shift from incidental to systematic recording in 2025 represents the most significant methodological change in the survey's history. Rather than recording species opportunistically during other survey work, each IOC week is now treated as a discrete sub-project with the explicit goal of maximising the number of species detected — not individual counts. Repeated passes through both savannah and riverine forest habitat within a single week consistently produce complementary species lists from each zone.

Mobile app recording. The Observation.org mobile app enabled photo-free submissions from late 2024 onward, removing a significant barrier for well-known species that are difficult to photograph (e.g. skulking understorey species, fast-flying raptors, owls heard but not seen). The Emerald-spotted Wood Dove is a concrete example: multiple app records in 2025 without photos were subsequently supported by photographic evidence, confirming that increased effort generates both more records and more photographic opportunities.

Rainfall and ecological context. The adjacent Blyderivier Botaniese Reservaat rainfall record (Johan Pienaar, 2000–present) documents the 2023/24 season at 491 mm — 15% below the long-term mean of 578 mm, with an unusually dry October–December 2023. While the direct impact on bird numbers is not quantifiable from current data, the concurrent decline in butterfly and Odonata richness in 2024 is consistent with a site-wide drought response. The subjective observation of reduced insect numbers in 2024 is relevant: insectivorous birds would be expected to respond to invertebrate prey availability.

Priorities for Future Seasons

- Continue weekly sub-projects throughout the survey window to build a robust multi-year presence/absence record with consistent effort
- Extend coverage into September–October — currently the most under-sampled period, likely productive for early-arriving summer species
- Target confirmed but infrequently recorded species (Pel's Fishing Owl, Crowned Eagle, Narina Trogon, Thick-billed Cuckoo) for repeated detection to confirm residency status
- Integrate rainfall and ERA5 temperature data with annual species richness to begin quantifying climate–biodiversity relationships
- Maintain photographic evidence for species not yet documented with photos at this site; 2025 systematic effort has already substantially increased the photographic record

6. Data Attribution & AI Acknowledgement

All bird observations were made by **Andries de Vries** (citizen scientist, Middelburg, Mpumalanga) at Swadini Forever Resort, Blyde River Canyon Nature Reserve. Records from 2019–2024 were submitted to the Virtual Museum (VMU) at the University of Cape Town and subsequently transferred to Observation.org with Accepted (with evidence) validation status. Records from December 2024 onward were submitted directly to Observation.org.

The rainfall context data used in this report was recorded by **Johan (Pine) Pienaar** of the Blyderivier Botaniese Reservaat, Mpumalanga — immediately adjacent to the study site — spanning August 2000 to April 2026.

This overview report was generated by **Claude**, an AI assistant developed by **Anthropic** (claude.ai), at the request of the observer in April 2026. Data processing, analysis, and report writing were performed by Claude; all biological interpretation and field knowledge were provided by the observer. The data remains the intellectual property of the observer and Observation.org.

Report generated: April 2026 · Data export: August 2025 · Total records: 806 · Species (normalised): 95 · Families: 42